

The monomial and its standard form

A number times some letters — the simplest object in algebra.

LESSON 65

1 × 40 MIN

MATEMATIKA 7, §29

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

What a monomial is	10 min
Standard form	12 min
Degree and similarity	12 min
Homework	6 min

LEARNING OBJECTIVES

- Define a monomial and recognise one.
- Write a monomial in standard form.
- Find the coefficient and the degree of a monomial.
- Recognise similar monomials.

TEXTBOOK

National	pp. 178–181
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

What a monomial is

A **monomial** is a product of numbers and letters, with natural-number exponents. No addition, no subtraction and no division by a letter.

EXPRESSION	A MONOMIAL?	WHY
$7a^2b$	yes	a product
$-x$	yes	$-1 \times x$
5	yes	a number alone
$a + b$	no	a sum
$\frac{3}{x}$	no	division by a letter
$\frac{x}{3}$	yes	$\frac{1}{3}x$

DIVIDING BY A NUMBER IS ALLOWED; DIVIDING BY A LETTER IS NOT

$\frac{x}{3}$ is the monomial $\frac{1}{3}x$; $\frac{3}{x}$ is not a monomial at all. The difference matters for everything that follows.

Standard form

A monomial is in **standard form** when it is written as one number followed by each letter once, in alphabetical order, with its exponent.

AS WRITTEN	STANDARD FORM
$3a \cdot 4b$	$12ab$
$x \cdot 5 \cdot x \cdot y$	$5x^2y$
$2b \cdot 3a \cdot a$	$6a^2b$
$-a \cdot a \cdot a \cdot 2$	$-2a^3$

MULTIPLY THE NUMBERS, COLLECT EACH LETTER

Two steps, always in that order. The alphabetical order is only a convention, but it makes two monomials easy to compare.

Degree and similarity

The **degree** of a monomial is the sum of the exponents of its letters. A number alone has degree 0.

MONOMIAL	COEFFICIENT	DEGREE
$7a^2b$	7	3
$-5x^3y^2$	-5	5
4	4	0
x	1	1

Two monomials are **similar** if their letter parts are identical — same letters, same exponents. Only similar monomials can be added.

PAIR	SIMILAR?
$3a^2b$ and $-7a^2b$	yes
$3a^2b$ and $3ab^2$	no
$5x$ and $5x^2$	no

A^2B AND AB^2 ARE NOT SIMILAR

Both have degree 3, but the letters carry different exponents. Similarity is about the letter part, not about the degree.

02

Worked examples

EXAMPLE 1 Write $2b \cdot 3a \cdot a$ in standard form, and give its coefficient and degree.

1

Numbers: $2 \times 3 = 6$.

2

Letters: $a \cdot a = a^2$, and one b .

3

$6a^2b$
Alphabetical order.

4

Coefficient 6, degree $2 + 1 = 3$.

ANSWER

$6a^2b$; 6 and 3

EXAMPLE 2 Which of $7a^2b$, $a + b$, $\frac{3}{x}$, $\frac{x}{3}$, 5 are monomials?

- ① $7a^2b$ — a product ✓
- ② $a + b$ — a sum ✗
- ③ $\frac{3}{x}$ — division by a letter ✗
- ④ $\frac{x}{3}$ and 5 ✓

ANSWER

$7a^2b$, $\frac{x}{3}$, 5

EXAMPLE 3 Are $3a^2b$ and $3ab^2$ similar? And $3a^2b$ and $-7a^2b$?

- ① Compare the letter parts.
- ② a^2b and ab^2 differ.
Not similar.
- ③ a^2b and a^2b agree.
- ④ Similar — only the coefficients differ.

ANSWER

No; yes

03 Interactive model

Write the same monomial six different ways on the board and ask which is “tidiest”; the class defines standard form before it is named.

This model could not start: den is not a function

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Monomial	Bir had	Одночлен
Standard form	Standart ko‘rinish	Стандартный вид
Coefficient	Koeffitsiyent	Коэффициент
Degree of a monomial	Bir hadning darajasi	Степень одночлена
Similar monomials	O‘xshash bir hadlar	Подобные одночлены
Letter part	Harfiy qism	Буквенная часть
Alphabetical order	Alifbo tartibi	Алфавитный порядок
Zero monomial	Nol bir had	Нулевой одночлен

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which is a monomial?

$$a + b$$

$$7a^2b$$

$$\frac{3}{x}$$

$$a - 1$$

Is $\frac{x}{3}$ a monomial?

yes

no

sometimes

only if $x > 0$

Standard form of $2b \cdot 3a \cdot a$:

$$6ab^2$$

$$6a^2b$$

$$5a^2b$$

$$6a^3$$

The degree of $-5x^3y^2$:

The degree of 4:

Are $3a^2b$ and $3ab^2$ similar?

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is $7a^2b$ a monomial?

ANSWER Yes

2. Is $a + b$ a monomial?

ANSWER No

3. Is $\frac{3}{x}$ a monomial?

ANSWER No

4. Is 5 a monomial?

ANSWER Yes

5. Coefficient of $7a^2b$

ANSWER 7

6. Degree of $7a^2b$

ANSWER 3

7. Degree of 4

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Standard form of $3a \cdot 4b$

ANSWER $12ab$

2. Standard form of $x \cdot 5 \cdot x \cdot y$

ANSWER $5x^2y$

3. Standard form of $2b \cdot 3a \cdot a$

ANSWER $6a^2b$

4. Coefficient and degree of $-5x^3y^2$

ANSWER -5 and 5

5. Are $3a^2b$ and $-7a^2b$ similar?

ANSWER Yes

6. Are $5x$ and $5x^2$ similar?

ANSWER No

7. Standard form of $-a \cdot a \cdot a \cdot 2$

ANSWER $-2a^3$

Hard · stretch and reason

7 PROBLEMS

1. Standard form of $(2a)^3 \cdot 3b$

ANSWER $24a^3b$

2. Degree of $(3x^2y)^2$

ANSWER 6

3. Standard form of $\frac{1}{2}x \cdot 4x^2y$

ANSWER $2x^3y$

4. Two similar monomials of degree 4

ANSWER $3a^2b^2$ and $-a^2b^2$

5. Is x^{-1} a monomial?

ANSWER No — the exponent is not natural

6. Degree of $7a^2b \cdot 3ab^3$

ANSWER 7

7. Coefficient of $-\frac{2x^3}{5}$

ANSWER $-\frac{2}{5}$

07 Homework

Homework — 5 tasks

Write every monomial in standard form before giving its coefficient or degree.

- Which of $4xy$, $x + 3$, $\frac{5}{y}$, $-2a^3$, 7 are monomials?
- Write $5a \cdot 2b \cdot a$ in standard form.
- Give the coefficient and degree of $-3x^2y^4$.

4. Are $4ab^2$ and $-ab^2$ similar? And $4ab^2$ and $4a^2b$?
5. Write $(3x)^2 \cdot 2y$ in standard form and give its degree.